

Monopolistic screening

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Lecture note + MSPC Chapter 4.3

Introduction

- We look at a problem of non-linear pricing by a monopolistic seller who faces a buyer with unknown valuation for her product.
- The seller sets the terms of the contract (the seller is the principal (P), the buyer is the agent (A)).
- The seller is a monopolist. Decides on price, quantity.

Model

- Risk-neutral **monopolist** with constant marginal costs c .
- The monopolist makes a **take-it-or-leave-it offer** consisting of a **transfer** T and a **quantity** q to a consumer (we assume there is only one) whose preferences for the product the monopolist cannot observe.
- The preference type θ of the consumer is either low or high, i.e. $\theta \in \{\theta_L, \theta_H\}$, $\theta_L < \theta_H$. Private information to the buyer.
- The monopolist only knows distribution, assigns probability β to the event that the consumer is of type θ_L .
- β is thus the belief of the principal: we can for now simply think of β as the “population share” of θ_L .

- A **consumer** of type θ derives utility of

$$u(q, T, \theta) = \theta v(q) - T$$

if he consumes $q \geq 0$ units of the monopolist's product and pays the monopolist a total of T dollars for these units.

- We do not worry about income or budget constraints. We simply assume that the agent can always pay T .
- If the consumer consumes nothing, his utility is 0. (His reservation utility $\underline{u} = 0$.)
- Assume that v is differentiable, $v(0) = 0$, $v'(\cdot) > 0$ and $v''(\cdot) < 0$ for all q .

- The profit of the **monopolist** is

$$\pi(q, T) = T - cq,$$

where $c \geq 0$ is the constant marginal cost of production.

Benchmark: First best

If there is no informational asymmetry, the principal can treat each type separately and offer a **type specific contract** that specifies a quantity and transfer (q_i, T_i) for each type:

$$q_L = q(\theta_L), \quad T_L = t(\theta_L) \text{ for type } \theta_L$$

$$q_H = q(\theta_H), \quad T_H = t(\theta_H) \text{ for type } \theta_H$$

Hence the principal solves the following problem for each type $\theta \in \{\theta_L, \theta_H\}$. For all $i = L, H$:

$$\max_{T_i, q_i} T_i - cq_i$$

subject to

$$\theta_i v(q_i) - T_i \geq \underline{u}$$

- In the optimal solution the participation constraint must hold with equality.
- So we can insert the binding constraint in the maximization problem:

$$\max_{q_i} \theta_i v(q_i) - cq_i$$

- The first best allocation is thereby characterized by

$$\theta_i v(q_i^{FB}) = T_i^{FB}$$

and the following first order condition:

$$\theta_i v'(q_i^{FB}) = c$$

- FOC implies that marginal utility of q equals the marginal costs from production. This is obviously efficient.

Comments

- Note that T_i^{FB} and q_i^{FB} are both functions of θ_i .
- $q_H^{FB} > q_L^{FB}$.
- $T_H^{FB} > T_L^{FB}$. This follows from: $\theta_i v(q_i^{FB}) = T_i^{FB}$.
- P 's expected profit is

$$\beta(T_L^{FB} - cq_L^{FB}) + (1 - \beta)(T_H^{FB} - cq_H^{FB})$$

- Both types of A get utility 0 (the reservation utility).

Comments

- The monopolist **extracts all the rents** from his customer.
- The optimal contract can be implemented in two different ways (both implying that every unit does not costs the same):
 - 1 The consumer has to pay at total price (transfer) T_i^{FB} for total quantity q_i^{FB} .
 - 2 Alternatively by type-specific **two-part tariffs**: The buyer is allowed to buy as much as he wants of the good at unit price c , provided that he pays a type-specific fixed fee equal to $\theta_i v(q_i^{FB}) - cq_i^{FB}$
 - The buyer chooses q_i so as to equalize marginal utility to marginal costs
- This is efficient! ... but why? It's a monopolist?

Sidestep: Traditional linear pricing - recap

- Contrast previous result to traditional linear pricing schedule (standard monopolistic pricing problem for each type).
- No entry fee, only a unit price (i.e. instead of $T(q) = F + Pq$ (two part tariffs) we now have $T(q) = Pq$ (linear pricing)).
- I.e., the contract specifies only a type specific unit price P_i and lets consumer choose q_i .
- Hence, a buyer of type θ_i chooses consumption to maximize

$$\theta_i v(q_i) - Pq_i$$

where q_i is the individual demand $q_i = D_i(P)$. The FOC implies the buyer chooses q_i , such that

$$\theta_i v'(q_i) = P \tag{1}$$

- From this we can derive the demand function $D_i(P)$ for each type.

Type i 's demand function:

The first order condition implicitly defines q_i as a function of P (implicit function theorem).

What is the slope of the demand function, $D_i(P)$?

Total differentiate (1)

$$\theta_i v''(q_i) \frac{dq_i(p)}{dp} = 1$$

$$\Rightarrow \frac{dq_i(p)}{dp} = D'_i(P^M) = \frac{1}{\theta_i v''(q_i)} < 0$$

For each type i , the monopolist solves

$$\max_{P_i} (P_i - c) D_i(P_i)$$

The optimal price is characterized by the first order condition

$$D_i(P_i^M) + D'_i(P_i^M) P_i^M = c D'_i(P_i^M) \Leftrightarrow P_i^M = c - \frac{D_i(P_i^M)}{D'_i(P_i^M)}$$

Hence, $P_i^M > c$. This results in inefficiently low consumption since $\theta_i v'(q_i^M) = P_i^M > c$.

Compare to the previous result. What is the difference?

Asymmetric information

Now, consider asymmetric information, so that the seller cannot observe and verify the type of a buyer.

What happens if she offers the two contracts designed for the first-best solution?

The type- θ_H buyer will actually choose the contract (q_L^{FB}, T_L^{FB}) since it gives him higher utility than 0, which he gets from (q_H^{FB}, T_H^{FB}) .

Indeed, since $\theta_H > \theta_L$

$$\theta_H v(q_L^{FB}) - T_L^{FB} > \theta_L v(q_L^{FB}) - T_L^{FB} = 0$$

- What contract should P offer to A under asymmetric information?
- At a first glance, this is a very difficult question, because there are many possible contracts and potential menus from which A could choose $\{(q_1, T_1), (q_2, T_2), \dots\}$.
- Fortunately, we can again rely on the “**Revelation Principle**”
 - We can restrict our attention to “truthful direct mechanisms”
 - A **direct mechanism** is a mechanism, in which the agent is asked to announce his type after it is realized. So A must announce either type θ_L or θ_H .
 - The mechanism specifies an outcome $(q(\hat{\theta}), T(\hat{\theta}))$ for each possible announcement $\hat{\theta}$.
 - A direct mechanism is **truthful** if it is incentive compatible for any type of the agent to announce his true type.

- Therefore, we again restrict attention to menus of contracts that specify for each type $\theta_i \in \{\theta_L, \theta_H\}$ a quantity q_i and a payment T_i .
- Let (q_L, T_L) be the contract for the L -type and (q_H, T_H) the contract for the H -type.
- P has to find contracts (q_L, T_L) and (q_H, T_H) that maximize her profit subject to the constraints that A trades with P and each type finds his contract weakly better than the other contract.

Therefore, P 's maximization problem is

$$\max_{(q_L, T_L), (q_H, T_H)} \beta(T_L - cq_L) + (1 - \beta)(T_H - cq_H)$$

subject to the following participation and incentive constraints.

$$\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L \quad (\text{ICH})$$

$$\theta_L v(q_L) - T_L \geq \theta_L v(q_H) - T_H \quad (\text{ICL})$$

$$\theta_H v(q_H) - T_H \geq 0 \quad (\text{PCH})$$

$$\theta_L v(q_L) - T_L \geq 0 \quad (\text{PCL})$$

- Instead of writing down the full Kuhn-Tucker problem we again ask which constraints bind and which do not.
- We show that the constrained maximization problem can be simplified so much that it becomes an unconstrained maximization problem (by inserting constraints we know are binding and removing non-binding constraints).

Step 1. (ICH) and (PCL) imply (PCH)

Proof

$$\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L \geq 0$$

- What is the intuition for this result?
- What is the implication of this result for type θ_H and for our maximization problem?

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$$\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L \geq 0$$

- What is the intuition for this result?
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 - The optimal contract gives type θ_H positive utility.

Step 1. (ICH) and (PCL) imply (PCH)

Proof

$$\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L \geq 0$$

- What is the intuition for this result?
- What is the implication of this result for type θ_H and for our maximization problem?
 - The optimal contract gives type θ_H positive utility.
 - We can disregard (PCH), since it is redundant.

Step 2. (PCL) must hold with equality at the optimum:

$$\theta_L v(q_L) - T_L = 0$$

Proof (Use (ICH)): Suppose it did not hold with equality at the optimum. Then, at the optimum

$$\theta_H v(q_H) - T_H \geq \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L > 0$$

and

$$\theta_L v(q_L) - T_L \geq \theta_L v(q_H) - T_H$$

- Now by increasing T_H and T_L by the same amount we do not change choices, and we increase P 's profit. This contradicts that we are at the optimum, where P 's profit is maximized.

Step 3. Since (PCH) can be ignored, we can show that (ICH) must hold with equality at the optimum.

Proof Suppose (ICH) doesn't hold with equality at optimum. Then

$$\theta_H v(q_H) - T_H > \theta_H v(q_L) - T_L > \theta_L v(q_L) - T_L = 0.$$

Then P could increase T_H by a sufficiently small ϵ without violating any constraint. This would increase her profit, contradicting that we are at an optimum to begin with.

- We therefore must have

$$\theta_H v(q_H) - T_H = \theta_H v(q_L) - T_L$$

This result implies that the H -type will be indifferent between the two contracts.

Illustration

- ① Participation constraint for low type binds: $\theta_L v(q_L) - T_L = 0$, i.e., type L 's indifference curve must pass through the origin.
- ② Fix a contract for the low type, (q_L, T_L) (must lie on the indifference curve from step 1)
- ③ Draw the high type's indifference curve through (q_L, t_L) (note the single crossing condition)
- ④ The contract for the high type cannot lie in:
 - ① Area A: here θ_L would choose (q_H, t_H) . That is, the low type's incentive constraint is violated
 - ② Area B: here θ_H would choose (q_L, t_L) . That is, the high type's incentive constraint is violated.
- ⑤ So (q_H, T_H) must lie in Area C.
 - ① This implies $q_L \leq q_H$
 - ② Note that (q_H, T_H) cannot lie in the interior of C. Then the monopolist could increase T_H without violating any constraints. So it must lie on the upper boundary of Area C.

Step 4. (ICL) does not bind at the optimum. I.e., (ICL) is slack.

Proof Use that (ICH) binds:

$$\theta_H(v(q_H) - v(q_L)) = T_H - T_L$$

Since $\theta_L < \theta_H$ we have

$$\theta_L(v(q_H) - v(q_L)) < T_H - T_L$$

or

$$\theta_L v(q_H) - T_H < \theta_L v(q_L) - T_L.$$

- See also figure based on last slide, from that it is clear that we expect (ICL) to not bind.
- Since (ICH) binds, it makes sense that (ICL) will not bind. Otherwise, the L -type would also be indifferent between the two contracts.

From (PCL) and (ICH) we get

$$T_L = \theta_L v(q_L) \quad (2)$$

and

$$\begin{aligned} T_H &= \theta_H(v(q_H) - v(q_L)) + T_L \\ &= \theta_H(v(q_H) - v(q_L)) + \theta_L v(q_L) \\ &= \theta_H v(q_H) - (\theta_H - \theta_L)v(q_L) \end{aligned} \quad (3)$$

The term $(\theta_H - \theta_L)v(q_L)$ is an “**information rent**.” It decreases the transfer T_H .

P has to give this utility to the H -type, otherwise the H -type chooses the same contract as the L -type.

Plugging the expressions for T_L and T_H in (2) and (3) into the original maximization problem yields the following unconstrained maximization problem:

$$\max_{q_L, q_H} \beta(\theta_L v(q_L) - cq_L) + (1 - \beta)[\theta_H v(q_H) - (\theta_H - \theta_L)v(q_L) - cq_H]$$

Interpretation:

The term in the first brackets is the full surplus generated from the purchases by type θ_L , which the principal appropriates entirely.

The term in square brackets is the full surplus generated from purchases by type θ_H minus his **information rent** $(\theta_H - \theta_L)v(q_L)$.

The latter represents the amount of surplus P must leave to type θ_H , if she does not want him to mimic the behaviour of type θ_L .

This informational rent is increasing in q_L .

There is a rent extraction vs. efficiency trade-off: lowering q_L lowers the information rent; but distorts allocation for the low type.

The first-order conditions are

$$\theta_H v'(q_H) = c$$

$$\theta_L v'(q_L) = \frac{c}{1 - \frac{1-\beta}{\beta} \frac{\theta_H - \theta_L}{\theta_L}} > c.$$

Compare these equations with the characterization of the first-best contracts.

In the optimum, the L -type gets a smaller quantity than under the first-best contract.

Comments

- The H -type gets the same quantity as under the first-best contract. This property also holds in general settings and is called “no distortion at the top”.
- The H -type (the potential deviator if the first best is implemented) gets positive (information) rent, while the L -type is held down to his reservation value.
- The L -type gets smaller quantity than under the first-best contract (the allocation is therefore inefficient). The intuition is that by reducing q_L the principal makes the contract for the L -type less attractive for the H -type and therefore lowers her information rent.
- The optimal menu of contracts depends on β . The smaller β , the smaller the quantity for the L -type.

Comparison with competitive screening

- We saw that with competitive screening there may be some eqm existence problems. This is tied to fierce competition among principals and as a result the market may break down.
- We do not see the same with monopolistic screening. The monopolist losses some profit compared to the FB but he still has positive profit and a contract solution exists.
 - So this is an illustration that with assymmetric information our standard complete markets results ala “competition is always a preferred enviroment” may not always hold